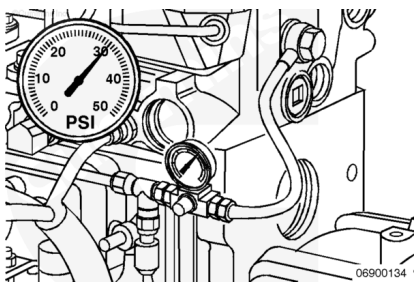


Test

⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.



If troubleshooting coolant in the fuel, fuel in the coolant, fuel in the oil, or oil in the fuel, pressurize the internal fuel drain line in the cylinder head and check for leaks.

Remove the fuel drain line at the rear of the cylinder head.
Refer to Procedure 006-013 in Section 6.
(/qs3/pubsys2/xml/en/procedures/216/216-006-013-tr.html)

If equipped with a quick disconnect fitting at the cylinder head, remove the check valve. Refer to Procedure 006-013 in Section 6. (/qs3/pubsys2/xml/en/procedures/216/216-006-013-tr.html)

Connect a regulated air supply hose to the cylinder head fuel drain port with a shutoff valve on the air supply side of the pressure gauge.

Apply air pressure.

Air Pressure		
kpa		psi
276	NOM	30

Shut off the air supply to the fuel drain port and monitor the pressure gauge reading. The pressure should hold steady. If the pressure drops rapidly, check for leaks around the:

- Test fittings
- Fuel connectors at the cylinder head.

Remove the rocker lever cover and check for air bubbles around the injectors. Refer to Procedure 003-011 in Section 3.
(/qs3/pubsys2/xml/en/procedures/218/218-003-011-tr.html)

Remove the radiator cap and check for air bubbles in the cooling system.

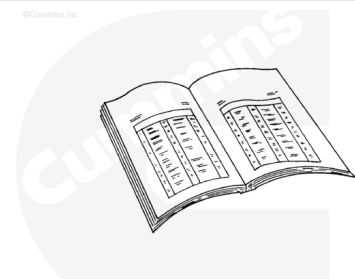
If the source of the leak can **not** be determined, remove the cylinder head and pressure test the complete cylinder head. Reference the Pressure Test section in this procedure. Replace the cylinder head, if necessary.

Preparatory Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

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ck800wa

⚠ WARNING ⚠

Coolant is toxic. Keep away from pets and children. If not reused, dispose of in accordance with local environmental regulations.

⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.

⚠ CAUTION ⚠

If removing the cylinder head with the injectors installed, be careful not to damage the tips of the injectors. Do not set the cylinder head down on the combustion face with the injectors installed. Damage to the injector tips will result.

- Disconnect the batteries. See the equipment manufacturer service information.
- Drain the coolant. Refer to Procedure 008-018 in Section 8. (/qs3/pubsys2/xml/en/procedures/220/220-008-018-

tr.html)

- Remove any original equipment manufacturer (OEM) accessories attached to the cylinder head. See the equipment manufacturer service information.
- Disconnect the air crossover connection. Refer to Procedure 010-019 in Section 10.
(/qs3/pubsys2/xml/en/procedures/220/220-010-019.html)
- Disconnect all water and heater hoses attached to the cylinder head. See the equipment manufacturer service information.
- Disconnect any engine wiring harness connections, as necessary.

Note : Omit the following steps if the engine is not equipped with the component or if it is not necessary to remove the component to remove the cylinder head. For some components, it is only necessary to remove the component if the cylinder head is being replaced or rebuilt.

- Remove the fuel filter and fuel filter head. Refer to Procedure 006-017 in Section 6.
(/qs3/pubsys2/xml/en/procedures/216/216-006-017-tr.html)
- Remove the fuel filter head bracket. Refer to Procedure 006-018 in Section 6.
(/qs3/pubsys2/xml/en/procedures/216/216-006-018-tr.html)
- Remove the fuel block connection. Refer to Procedure 006-021 in Section 6.
(/qs3/pubsys2/xml/en/procedures/216/216-006-021-tr.html)
- Remove the air intake manifold. Refer to Procedure 010-023 in Section 10.
(/qs3/pubsys2/xml/en/procedures/216/216-010-023-tr.html)
- Remove the drive belt. Refer to Procedure 008-002 in Section 8. (/qs3/pubsys2/xml/en/procedures/220/220-008-002.html)
- Remove the fan hub pulley. Refer to Procedure 008-039 in Section 8. (/qs3/pubsys2/xml/en/procedures/216/216-008-039-tr.html)
- Remove the fan hub assembly. Refer to Procedure 008-036 in Section 8.
(/qs3/pubsys2/xml/en/procedures/216/216-008-036-tr.html)

- Loosen the alternator link, mounting bolt, and water inlet connection capscrews. Remove the alternator bracket mounting capscrews, and pivot the alternator away from the engine. Refer to Procedure 013-001 in Section 13. (</qs3/pubsys2/xml/en/procedures/216/216-013-001-tr.html>)
- Remove the alternator bracket from the thermostat housing. Refer to Procedure 013-003 in Section 13. (</qs3/pubsys2/xml/en/procedures/216/216-013-003-tr.html>)
- Remove the thermostat housing and thermostat from the engine. Refer to Procedure 008-013 in Section 8. (</qs3/pubsys2/xml/en/procedures/216/216-008-013-tr.html>)
- Remove the coolant temperature sensor. Refer to Procedure 019-019 in Section 19. (</qs3/pubsys2/xml/en/procedures/216/216-019-019.html>)

Note : The following components are required to be removed in order to remove the cylinder head.

- Remove the turbocharger. Refer to Procedure 010-033 in Section 10. (</qs3/pubsys2/xml/en/procedures/216/216-010-033-tr.html>)
- Remove the exhaust manifold. Refer to Procedure 011-007 in Section 11. (</qs3/pubsys2/xml/en/procedures/216/216-011-007-tr.html>)
- Remove the rocker lever cover. Refer to Procedure 003-011 in Section 3. (</qs3/pubsys2/xml/en/procedures/216/216-003-011-tr.html>)
- Remove the rocker lever housing. Refer to Procedure 003-013 in Section 3. (</qs3/pubsys2/xml/en/procedures/216/216-003-013-tr.html>)
- Remove the rocker levers. Refer to Procedure 003-008 in Section 3. (</qs3/pubsys2/xml/en/procedures/220/220-003-008-tr.html>)
- Remove the crossheads. Refer to Procedure 002-001 in Section 2. (</qs3/pubsys2/xml/en/procedures/220/220-002-001-tr.html>)
- Remove the push rods. Refer to Procedure 004-014 in Section 4. (</qs3/pubsys2/xml/en/procedures/220/220-004-014-tr.html>)

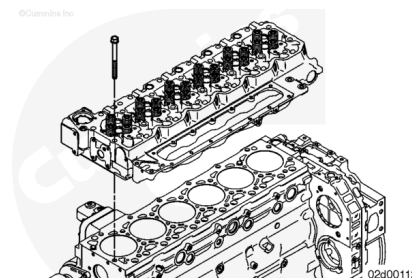
- Remove the fuel drain lines. Refer to Procedure 006-013 in Section 6. (</qs3/pubsys2/xml/en/procedures/216/216-006-013-tr.html>)
- Remove the injector supply lines. Refer to Procedure 006-051 in Section 6. (</qs3/pubsys2/xml/en/procedures/216/216-006-051-tr.html>)
- Disconnect the fuel rail from the cylinder head. Refer to Procedure 006-060 in Section 6. (</qs3/pubsys2/xml/en/procedures/216/216-006-060-tr.html>)
- Remove the fuel connectors. Refer to Procedure 006-052 in Section 6. (</qs3/pubsys2/xml/en/procedures/216/216-006-052-tr.html>)

Note : Do not remove the injectors at this time. Remove the cylinder head with the injectors installed so the injector protrusion can be checked.

Remove

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To reduce the possibility of serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.

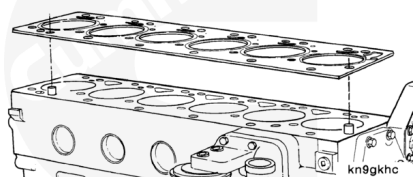


⚠ CAUTION ⚠

If removing the cylinder head with the injectors installed, be careful not to damage the tips of the injectors. Do not set the cylinder head down on the combustion face with the injectors installed. Damage to the injector tips will result.

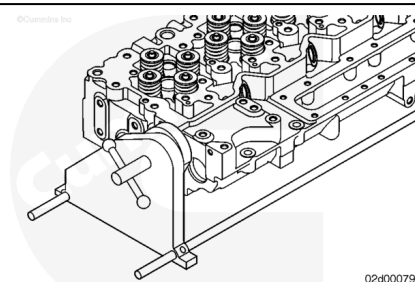
Remove the cylinder head capscrews and cylinder head.

Remove the cylinder head gasket from the cylinder block.

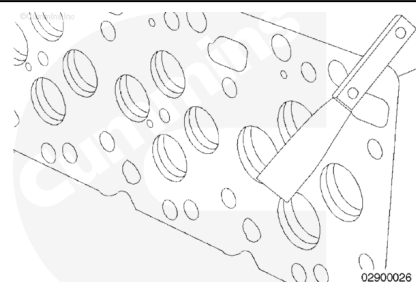


Initial Check

Install the cylinder head in the Cummins® cylinder head holding fixture, Part Number ST-583.

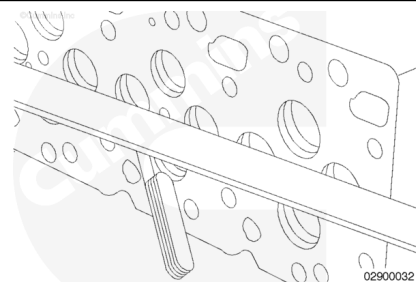


Scrape the gasket material and clean the combustion deck surfaces on the cylinder block and cylinder head.



Use a straightedge and a feeler gauge to inspect the cylinder head combustion surface for flatness.

Cylinder Head Flatness				
	mm		in	
End-to-End		0.305	MAX	0.012
Side-to-Side		0.076	MAX	0.003



If out of specification, determine if the cylinder head can be resurfaced, or if the cylinder head **must** be replaced by:

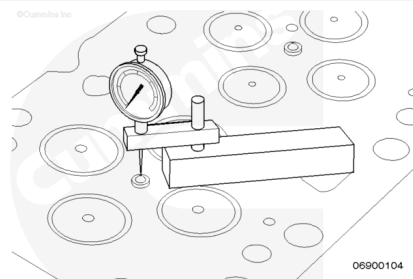
- Measuring and recording valve depth
- Measuring and recording injector protrusion.

If valve depth and injector protrusion specifications can be maintained, the cylinder head can be resurfaced. If the specifications can **not** be maintained, the cylinder head **must** be replaced.

Install a Cummins® depth gauge assembly, Part Number 3164438, on the cylinder head combustion deck and zero the gauge.

Rotate the depth gauge to measure the injector protrusion at the highest point on the injector.

Record the injector protrusion for each injector.



Injector Protrusion		
mm		in
2.45	MIN	0.096
3.15	MAX	0.124

Note : Do not use thicker or double stacked injector sealing washers to correct injector protrusion. This will cause misalignment of the high-pressure fuel connector.

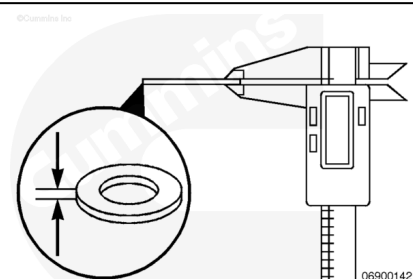
If the injector protrusion is out of specification, check the thickness of the injector sealing washer. Refer to Procedure 006-026 in Section 6.

(/qs3/pubsys2/xml/en/procedures/216/216-006-026-tr.html)

If the sealing washer is the correct thickness, check to make sure the injector bore is clean and free of debris. Also make sure that sealing washers are **not** 'stacked' in the injector bore.

If the injector protrusion is within specification, remove the injectors. Refer to Procedure 006-026 in Section 6.

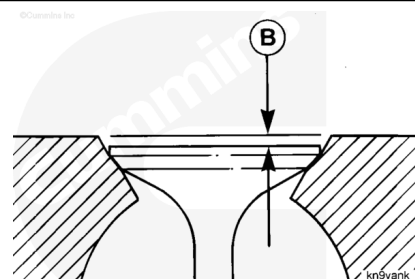
(/qs3/pubsys2/xml/en/procedures/216/216-006-026-tr.html)



Install a Cummins® depth gauge assembly, Part Number 3164438, on the cylinder head combustion deck and zero the gauge.

Rotate the depth gauge to measure the valve recession into the cylinder head (B).

Record the valve depth for each valve.

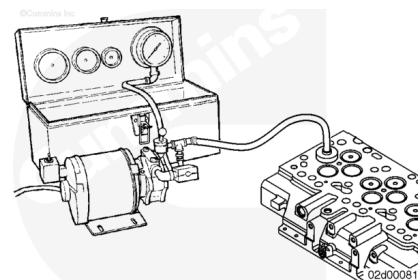


Intake Valve Depth (Installed)		
mm		in
0.584	MIN	0.023
1.092	MAX	0.043

Exhaust Valve Depth (Installed)		
mm		in
0.965	MIN	0.038
1.473	MAX	0.058

Note : Valve depth can be increased slightly for resurfacing of the cylinder head by lapping the valves.

If a leaking valve is suspected or if the cylinder head was recently rebuilt, vacuum test the valves and valve seats. Use Cummins® valve vacuum tester, Part Number 3824277, and Cummins® cup, Part Number ST-1257-6. The vacuum **must not** drop more than 25.4 mm Hg [1.0 in Hg] in five seconds.



Note : If a vacuum tester is not available, with the valve removed, use a lead pencil or Dykem™ to mark across the valve face. Install the valve in the valve guide. Hold the valve against the valve seat, and rotate the valve backward and forward three or four times. Correct contact against the valve seat will break the marks on the valve face.

Valve to Valve Seat Vacuum			
	kpa		in-hg
Used	51	NOM	15
New	69	NOM	20

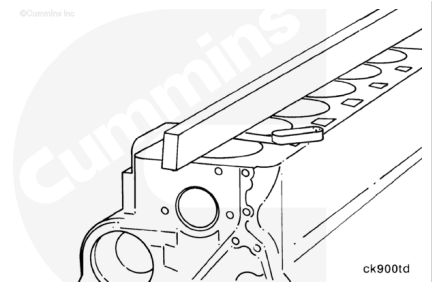
If out of specification, disassemble the cylinder head and inspect for damaged valves and/or valve seats. Repair as necessary by:

- Cleaning the valve/valve seat and lapping the valves
- Replacing the damaged valve/valve seat, if available
- Replacing the cylinder head.

Use a straightedge and feeler gauge to measure the overall flatness of the cylinder block.

Cylinder Block Flatness	
-------------------------	--

	mm		in	
End-To-End		0.075	MAX	0.003
Side-To-Side		0.075	MAX	0.002



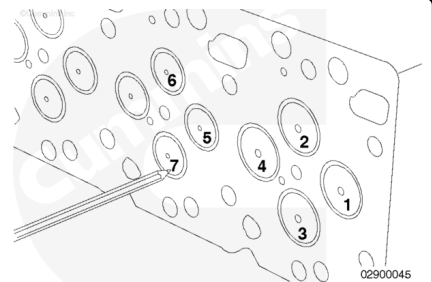
Inspect the combustion deck for any localized dips or imperfections.

If out of specification, determine if the cylinder block can be resurfaced or if the cylinder block **must** be replaced. Refer to Procedure 001-026 in Section 1.

(/qs3/pubsys2/xml/en/procedures/216/216-001-026.html)

Disassemble

Mark the valves to identify their location.

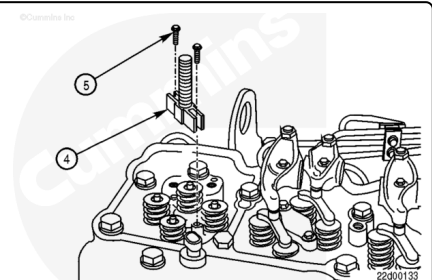


Compress the valve springs using the Cummins® valve spring compressor service tool, Part Number 3164329.

Position the replacer screw (4) above the injector bore and install the two capscrews (5) in the cylinder head where the hold-down clamp screws were removed.

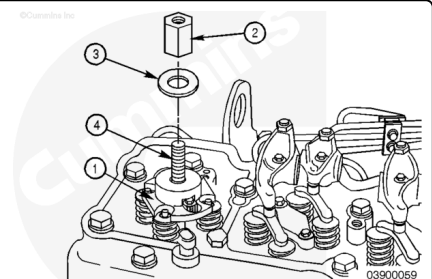
Tighten the capscrews (5).

Torque Value: 5 n•m [44 in-lb]



Note : The valves are **not** evenly spaced from the injector bore. It is important to align the slots in the valve spring compressor plate with the valve springs.

Apply anti-seize compound to the replacer screw (4) threads. Always read and follow label precautions.



Position the valve spring compressor plate (1) on the replacer screw (4) and align the slots in the valve spring compressor plate with the valve springs.

Install the washer (3) and nut (2) on the replacer screw (4).

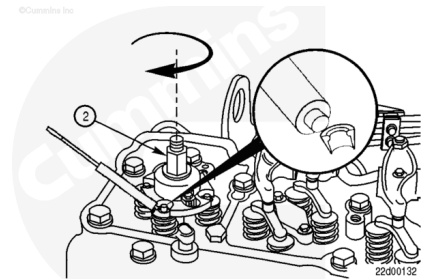
⚠ WARNING ⚠

Valve springs are under tension and can act as projectiles if released. To reduce the possibility of eye injury, wear safety glasses with side shields.

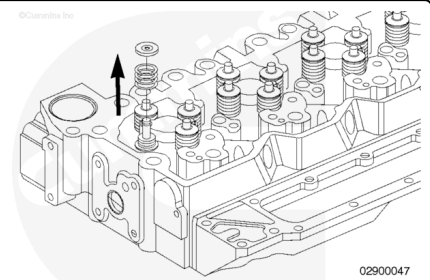
Turn the nut (2) **clockwise** to compress the valve springs.

Continue turning the nut (2) **clockwise** until the valve collets can be removed using a magnetic tool, such as the end of a magnetic screwdriver.

Remove the valve collets and the valve spring compressor service tool.

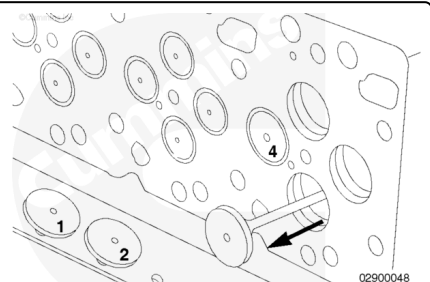


Remove the valve spring retainer and valve springs.



Note : Keep the valves in a labeled rack with the associated valve collets, spring retainers, and springs. This will aid in the assembly of the components as a matched set.

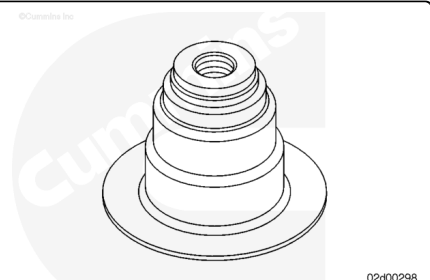
Remove the valves.



Note : Prior to removal of the valve stem seals, note the color of the valve stem seal installed at each valve location. The same color valve stem seal must be installed in the same valve location when the cylinder head is assembled.

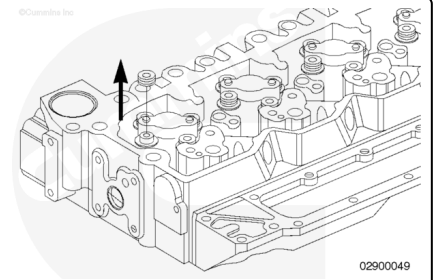
Colors Used:

- Green (exhaust valves)

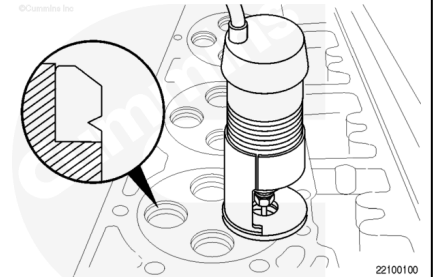


- Yellow (intake and exhaust valves).

Use Cummins® boot pliers, Part Number 3163293, to remove the valve stem seals.



Note : Prior to the removal of the valve seat inserts, reference the Initial Check and Clean and Inspect for Reuse sections in this procedure. The condition of the valve, the amount of recess, and the sealing of the valve on the seat insert all help determine whether or not a seat insert needs to be replaced.



- If required, remove the valve seat inserts.
- Inspect the valve-insert-to-cylinder-head contact area. A sufficient groove for the remover **must** exist.
- If there is sufficient valve insert groove area, proceed to the next step.
- If the valve insert groove area is **not** sufficient, use the Cummins® Valve Seat Insert Cutting Kit, Part Number 3376405, to create a sufficient groove.

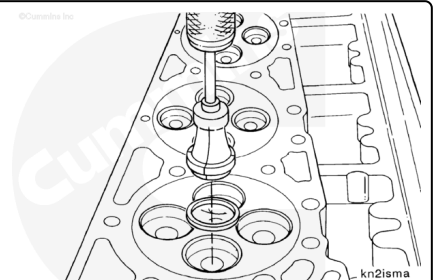
Use the Cummins® slide hammer remover, Part Number 3376617, with Cummins® valve insert remover, Part Number 3165170, to remove the valve seats.

Note : Make certain the valve insert remover assembly is perpendicular to the cylinder head when installed.

Insert the valve insert remover assembly into the valve insert and rotate the T-handle **clockwise** until the remover loosely grips the valve insert.

Position the valve insert remover assembly into the valve insert groove area. Tighten the T-handle firmly, allowing the remover to expand under the valve insert or into the cut groove.

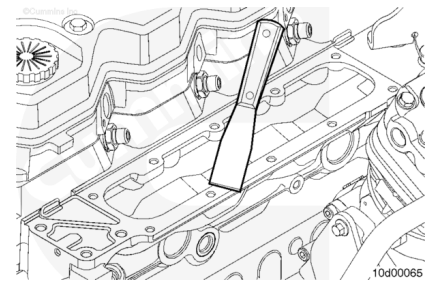
Strike the slide hammer remover against the top nut until the valve insert is removed. Turn the T-handle **counterclockwise** to release the valve insert from the remover.



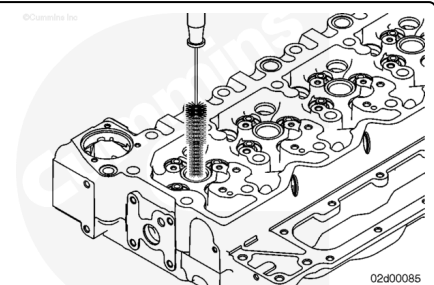
Clean and Inspect for Reuse

Keep the gasket material, and any other foreign materials, out of the air intake.

If removed, clean the cylinder head sealing surfaces where the air intake manifold seals.



Use an injector bore brush, Part Number 3822510, or equivalent, to clean the carbon from the injector seat.

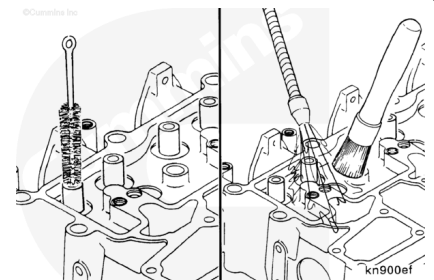


⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Use a bristle brush to clean the inside diameter of the valve guide bore.

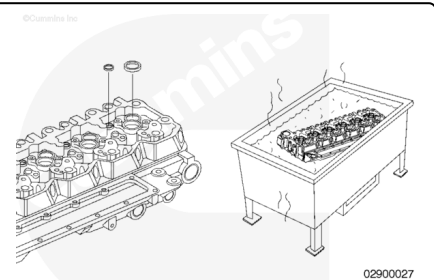
Blow out with compressed air.



Excessive deposits can be cleaned in an acid tank, but the expansion plugs **must** be removed first. Refer to Procedure 017-002 in Section 17.

(/qs3/pubsys2/xml/en/procedures/220/220-017-002.html)

If required, clean the buildup of deposits in the coolant passages.



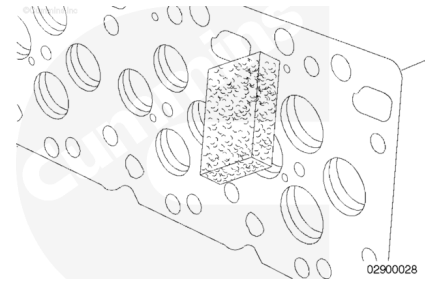
⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the

possibility of personal injury.

⚠ WARNING ⚠

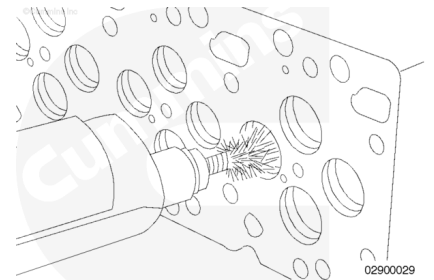
Some solvents are flammable and toxic. Read the manufacturer's instructions before using.



Clean the cylinder head combustion deck with Cummins® abrasive hand pad, Part Number 3823258, or equivalent, and solvent.

⚠ WARNING ⚠

Wear protective eye covering while cleaning carbon deposits to reduce the possibility of personal injury.



⚠ CAUTION ⚠

Contacting the valve seat with the wire wheel while it is spinning will damage the valve seat. If this occurs, new valve seats must be cut or new valve seat inserts must be installed.

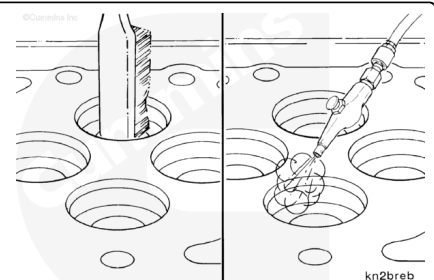
Inspect the area within 1/8-inch of the firing ring diameter. Any wear that can be felt with a fingernail within the 1/8-inch area is unacceptable, making the cylinder head **not** reusable. Wear beyond this 1/8-inch area will have no effect on future combustion sealing and the usability of the cylinder head.

Note : An inferior-quality wire wheel will lose steel bristles during operation, causing additional contamination.

Clean carbon deposits from the valve pockets with a high-quality steel wire wheel installed in a drill or die grinder.

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



⚠ WARNING ⚠

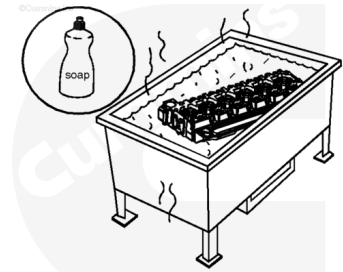
Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Use a wire brush and solvent to clean the deposits from the valve seat insert bores if it was necessary to remove the valve seat inserts.

Dry with compressed air.

⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.



Wash the cylinder head in a hot, soapy water solution.

Rinse the cylinder head with clean water.

Dry the cylinder head with compressed air.

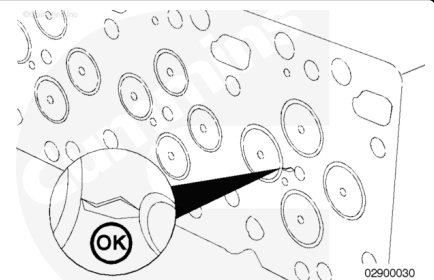
Cylinder Head Cracks - Reuse Guidelines

With the cylinder head cleaned, inspect the cylinder head for signs of cracks, fretting, and discoloration that would prohibit reuse.

To help identify cracks in the cylinder block, use the Cummins® Crack Detection Kit, Part Number 3375432.

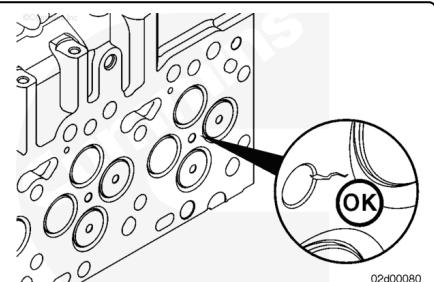
Pay close attention to areas of the cylinder head that include:

- Injector bore
- Combustion face
- Valve seats
- Valve guides.



The reuse guidelines for a cylinder head with a crack extending from the injector bore to the intake valve seat are as follows:

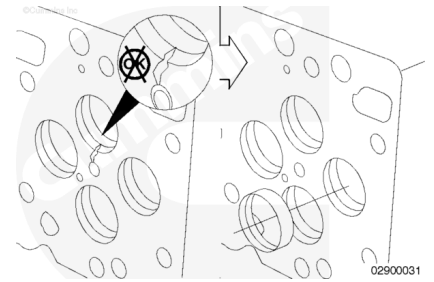
- If the crack does **not** extend into the valve seat, the cylinder head is reusable.



⚠ CAUTION ⚠

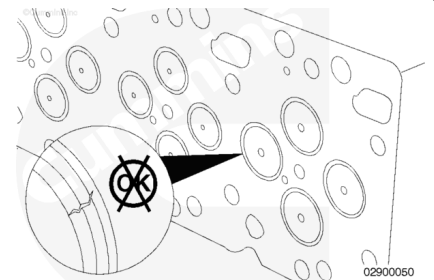
Failure to replace the cylinder head for a crack that extends into or through the valve seat bore will result in a valve seat insert falling out. Engine damage will result.

- If the crack extends into or through the valve seat, the cylinder head **must** be replaced.



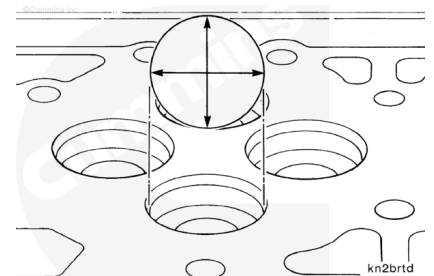
If still installed, inspect the valve seats for cracks or burnt spots.

If the valve seat inserts are damaged, the cylinder head **must** be machined or replaced.



If the valve seat insert was removed in the Disassemble section, measure the inside diameter of the valve seat insert bore in the cylinder head.

Cylinder Head Insert Bore Inside Diameter.		
mm		in
34.847	MIN	1.3719
34.863	MAX	1.3726



Note : Before cutting the cylinder head, verify valve seat inserts are available for the engine being serviced. If none are available, the cylinder head **must** be replaced.

If out of specification, the valve seat insert bore can be oversized 0.254 mm (0.010 in) or 0.508 mm (0.020 in).

Use Cummins® valve seat cutter, Part Number ST257, or Cummins® valve seat insert tool kit with Cummins® valve guide arbor, Part Number 3165184, to cut the cylinder head to accept oversize valve seat inserts. Use Cummins® valve seat cutter, Part Number 3165183, for 0.254 mm (0.010 in) or 3165184 for 0.508 mm (0.020 in).

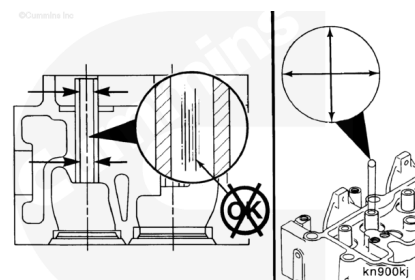
Valve Guide - Reuse Guidelines

Inspect the valve guides for scuffing or scoring.

Measure the valve guide inside diameter (I.D.)

Valve Guide Bore Diameter

mm		in
7.027	MIN	0.2767
7.077	MAX	0.2786



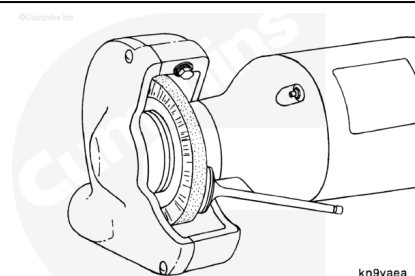
If the valve guide bore is worn larger than the maximum specified, or if inspection reveals damaged valve guides, the cylinder head **must** be replaced.

⚠ WARNING ⚠

Wear protective eye covering when cleaning the valves with a wire wheel to reduce the possibility of personal injury.

Clean the valve heads with a soft wire wheel.

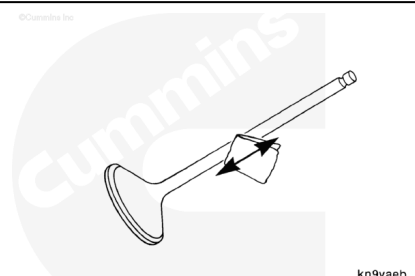
Note : Keep the valves in a labeled rack to prevent mixing. This will aid in analysis and assembly.



⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.

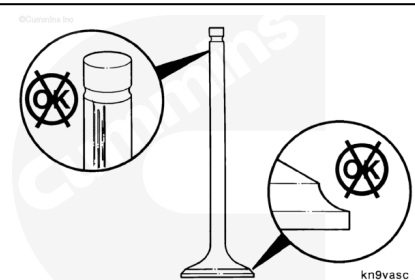
Polish the valve stems with a Cummins® abrasive pad, Part Number 3823258, and solvent.



Valve - Reuse Guidelines

Inspect the valves for:

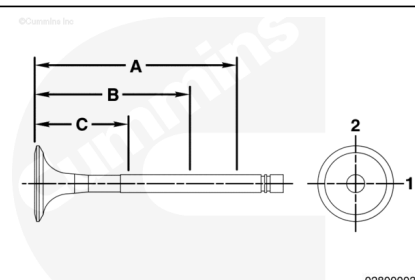
- Excessive wear on the heads and stems
- Excessive wear on the valve stem tip
- Bends and distortion.



Inspect the valves for damage and the collet grooves for wear.

Measure the outside diameter of the valve stem.

Three measurements **must** be taken of each valve stem at 45 mm [1.77 in.], 70 mm [2.75 in.], 95 mm [3.74 in.] from the surface of the valve head.

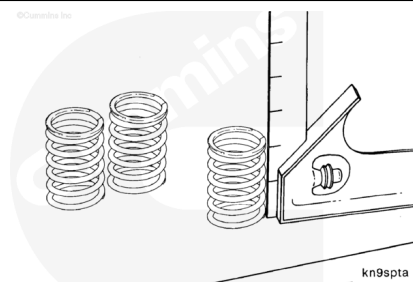


Valve Stem Outside Diameter

mm		in
6.99	MIN	0.2752
7.01	MAX	0.2760

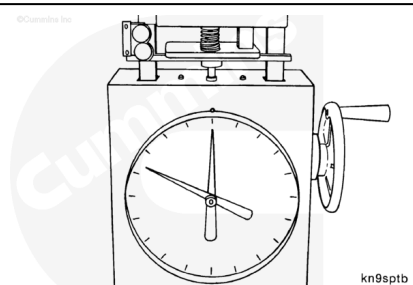
If the valves are damaged or the stems are worn smaller than the minimum specified, the valves **must** be replaced.

Inspect the valve springs.



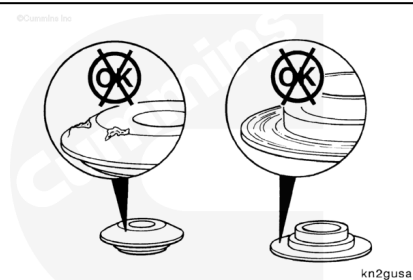
Use Cummins® valve spring tester, Part Number 3375182, to compress the valve spring. A load of 320.8 to 358.8 N [72 to 80.7 lbf] is required to compress a spring to a height of 35.33 mm [1.39 in].

Note : If the valve spring is not within specification, a new valve spring must be used.



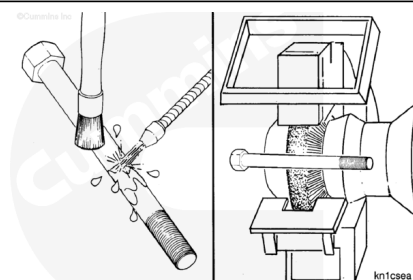
Inspect the valve spring retainers and valve collets for damage or worn areas.

Discard and replace damaged or worn parts.



⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



⚠ WARNING ⚠

Some solvents are flammable and toxic. Read the manufacturer's instructions before using.

⚠ CAUTION ⚠

Do not use caustic or acid solutions to clean the cylinder head capscrews. Component damage can occur.

Use a petroleum-based solvent to clean the capscrews.

Clean the capscrews thoroughly with a wire brush, soft wire wheel, or nonabrasive bead blast to remove deposits from the shank and threads.

Cylinder Head Capscrew - Reuse Guidelines

Inspect the cylinder head capscrews for damaged threads, corroded surfaces, or a reduced diameter (due to capscrew stretching).

Do **not** reuse cylinder head capscrews under the following conditions:

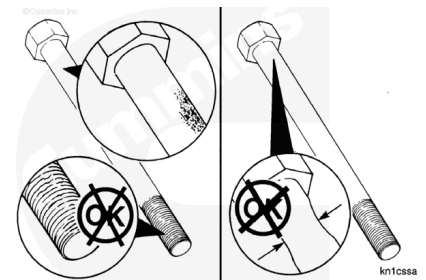
- Visible corrosion or pitting exceeding 1 sq cm [0.155 sq in] in area.

Example:

Acceptable is 9.525 x 9.525 mm [3/8 x 3/8 in].

Unacceptable is 12.700 x 12.700 mm [1/2 x 1/2 in].

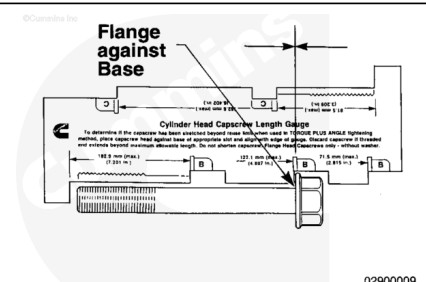
- Visible corrosion or pitting exceeds 0.12 mm [0.005 in] in depth.
- Visible corrosion or pitting is located within 3.2 mm [1/8 in] of the fillet or thread.
- Stretched beyond "free-length" maximum. See the measurement procedure below:



Free-Length Measurement

Note : If the capscrews are not damaged, they can be reused throughout the life of the engine, unless the specified free length is exceeded.

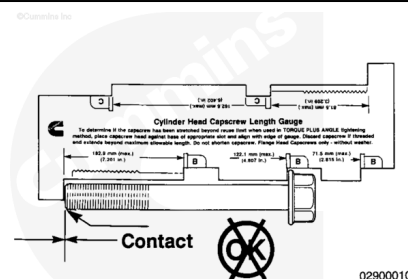
To check the capscrew free length using Cummins® capscrew length gauge, Part Number 3164057, place the head of the capscrew in the appropriate slot, with the flange against the base of the slot.



Note : Most new cylinder head gaskets and upper engine gaskets include the Cummins® capscrew length gauge, Part Number 3164057.

If the end of the capscrew touches the foot of the gauge, the capscrew is too long and **must** be discarded.

Capscrew Free Length		
mm		in
152.1	MAX	5.99

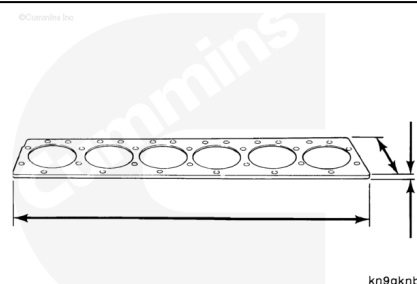


Pressure Test

If troubleshooting an internal coolant leak or coolant loss symptom, a cylinder head test fixture can be fabricated from a flat piece of steel or aluminum to pressure test the cylinder head.

Reference the following table for test fixture dimensions.

16 mm	Thickness	0.630 in
749 mm	Length	29.5 in
193 mm	Width	7.6 in



Note : Use the old cylinder head gasket as a pattern for drilling the capscrew holes.

Install the thermostat, water outlet connection, and mounting capscrews.

Note : If equipped with remote mounted thermostat, refer to the OEM service manual for the instructions that apply to the engine being serviced.

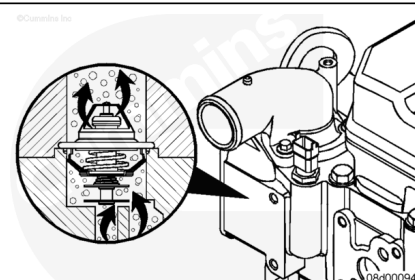
Tighten the capscrews

Torque Value: 10 n•m [89 in-lb]

Install the engine coolant temperature sensor, located next to the water outlet connection.

Torque Value: 18 n•m [159 in-lb]

Note : The thermostat contains two check balls to vent air past the thermostat when closed. Install a rubber

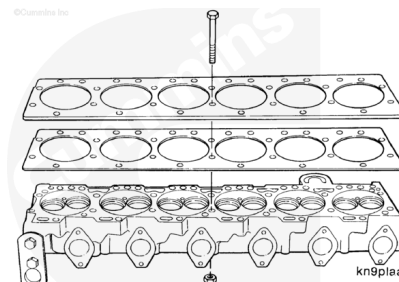


cap and hose clamp over the water outlet connection to prevent air from leaking through the check balls.

Install the cylinder head water test fixture.

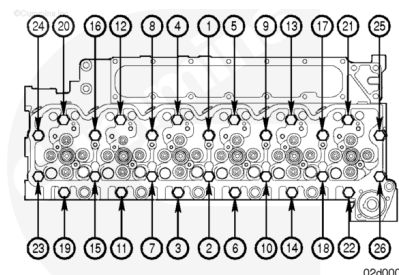
- Install a new head gasket.
- Install the test plate.
- Install the following:
 - 26 - 180-mm-long grade 12.9 flange head capscrews
 - 26 - M12 x 1.75 hex flange nuts
 - 52 - 12-mm washers.

Note : Place a washer between each capscrew and the head, and between each nut and test plate. This will prevent mutilation on the surface of the cylinder head.

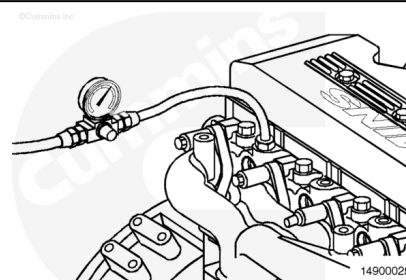


Tighten the nuts in the sequence illustrated.

Torque Value: 80 N·m [59 ft-lb]

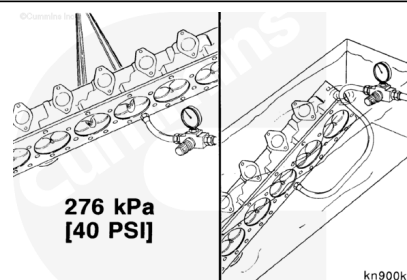


Service Tip: To apply air pressure to the cylinder head, remove one of the pipe plugs located on the exhaust side of the cylinder head. This is the same port used, when the cylinder head is installed to check cylinder block coolant pressures.



⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To reduce the possibility of serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



Note : Make sure to plug or seal any open coolant ports before pressure testing the cylinder head.

Connect a Cummins® regulated air supply hose, Part Number 3164231, to the cylinder head.

Apply air pressure.

Measurements

	kpa	psi
Air Pressure:	276	40

Use a nylon lifting strap and a hoist to place the cylinder head in a tank of heated water.

Measurements

	celsius	fahrenheit
Water Temperature:	60	140

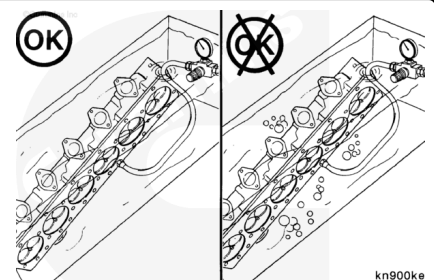
Note : The cylinder head must be completely submerged in the water.

Inspect the head. Bubbles indicate an air leak.

If any bubbles exist, verify the air leak is **not** coming from:

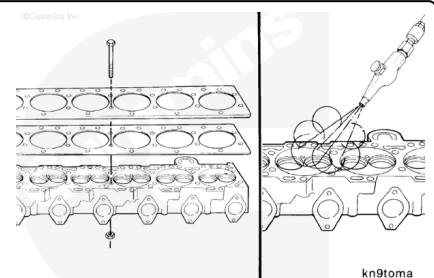
- Any cup plugs or fittings installed in the cylinder head
- Test fixture or air line fittings
- Water outlet connection.

If the above checks out OK and bubbles are present, the cylinder head leaks and it **must** be replaced.



⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

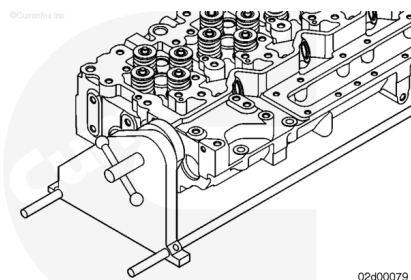


Remove the test fixture.

Use compressed air to dry the cylinder head.

Assemble

Install the cylinder head in the Cummins® cylinder head holding fixture, Part Number ST-583.

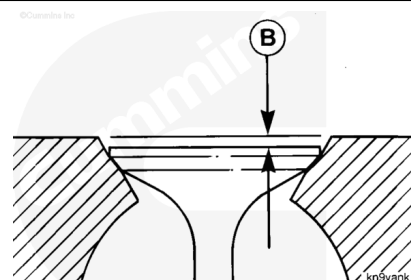


When installing the valve seat inserts, note that the exhaust and intake valve seat inserts are **not** the same.

Valve seat angle:

- Intake 30-degrees
- Exhaust 45-degrees.

If new valve seat inserts are installed, check valve depth and perform a valve leak test. Reference the Initial Check section in this procedure.

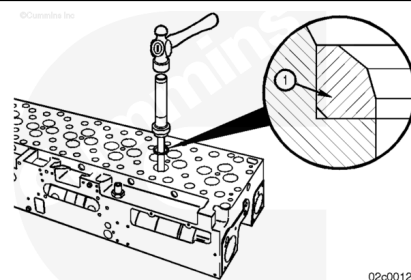


If the valve seat inserts were removed in the Disassemble section, new inserts **must** be installed.

The insert chamfer (1) **must** be installed toward the bottom of the counterbore.

Use Cummins® valve seat installer, Part Number 3165171, to drive the intake and exhaust valve seat inserts into the counterbore.

Use a dead blow hammer with the seat drivers to install the new valve seat inserts.

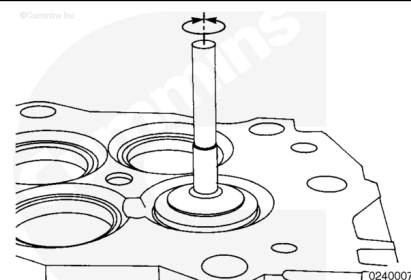


If new valve seat inserts were installed and/or the valve leakage was above specification, the valve seat/valve can be lapped.

Lubricate the stems with Society of Automotive Engineers (SAE) 15W-40 engine oil before installing the valves.

Use a Cummins® fine lapping compound, Part Number 3375805, or equivalent to apply a thin and even coating on the valve face.

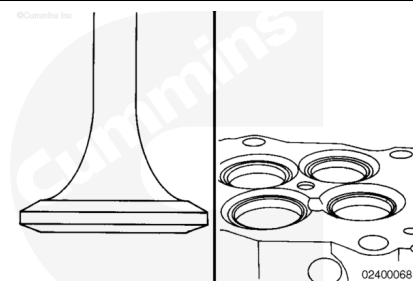
Use a power or a hand suction lapping tool to provide pressure in the center of the valve.



Turn the valve backward and forward. Continue lapping until the compound shows a continuous contact pattern on both the valve seat insert and the valve.

⚠ CAUTION ⚠

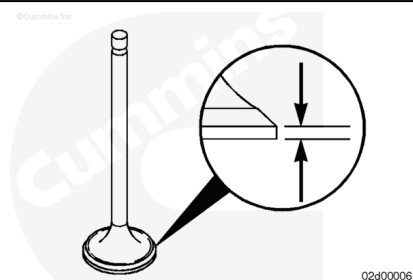
Lapping compound is an abrasive material. Failure will result if the cylinder head, the valves, and the valve seats are not cleaned thoroughly.



Clean the lapping compound from the parts.

If lapping of the valves was required, measure the rim thickness to determine if there is enough rim material left.

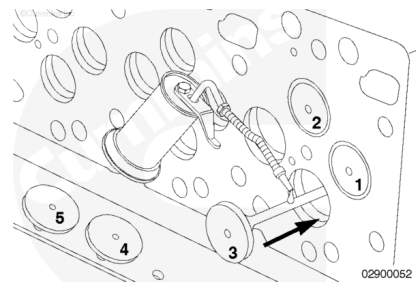
Valve Rim Thickness Limit		
mm		in
0.79	MIN	0.031



If the valve rim thickness is **not** within the limits, a new valve **must** be used.

⚠ CAUTION ⚠

Lubricate all the valve guide bores and valve stems with SAE 15W-40 engine oil. Failure to lubricate the valve guides and valve stems can result in premature valve guide wear.



Lubricate the valve stems with clean SAE 15W-40 engine oil before installing the valves.

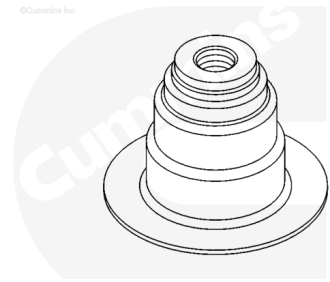
Install the valves.

Note : If the same valves as previously removed are to be used, make sure to install the valves in the same locations from which the valves were removed.

If the cylinder head will **not** be used immediately, lubricate the valve stems with Cummins® assembly lubricant, Part Number 3163087, or equivalent.

⚠ CAUTION ⚠

The same color valve stem seal must be installed in the same location as removed. Incorrect valve stem seals will result in excessive oil consumption and internal engine damage.



Install new valve stem seals of the same color as removed, and in the same locations.

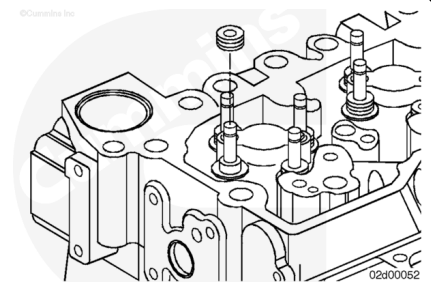
There are two colors of valve stem seals used:

- Green (exhaust valves)
- Yellow (intake and exhaust valves).

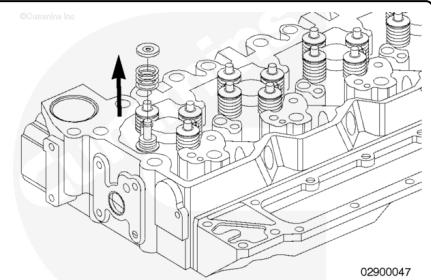
Use a Cummins® valve stem installation tool, Part Number 3164055, to install new valve stem seals.

The valve stem seals can be installed by hand. The installation tool will aid with installing the valve stem seals, but is **not** mandatory.

Use hand pressure to keep the valves from falling out during installation.



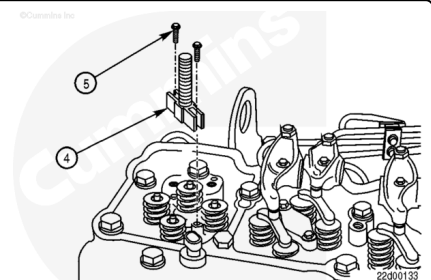
Install the valve spring retainer and valve springs.



Compress the valve springs using the Cummins® valve spring compressor service tool, Part Number 3164329.

Position the replacer screw (4) above the injector bore and install the two capscrews (5) in the cylinder head where the hold-down clamp screws were removed.

Tighten the capscrews (5).



Torque Value: 5 n·m [44 in-lb]

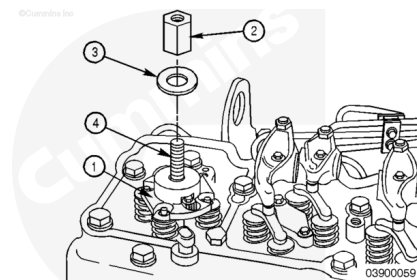
The valves are **not** evenly spaced from the injector bore. It is important to align the slots in the valve spring compressor plate with the valve springs.

Apply anti-seize lubricant to the replacer screw (4) threads.

Always read and follow label precautions.

Position the valve spring compressor plate (1) on the replacer screw (4) and align the slots in the valve spring compressor plate with the valve springs.

Install the washer (3) and nut (2) on the replacer screw (4).



⚠ WARNING ⚠

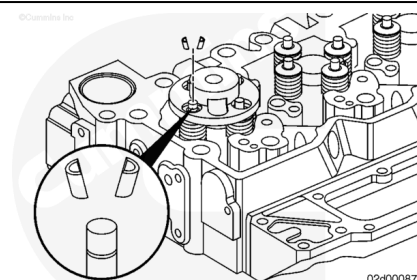
Valve springs are under tension and can act as projectiles if released. To reduce the possibility of eye injury, wear safety glasses with side shields.

Compress the valve springs until the valve collets can be installed.

Install the valve collets.

Service Tip: Use Cummins® assembly lube, Part Number 3163087, or equivalent, on the valve collets to help hold them in place until the valve spring compressor is released.

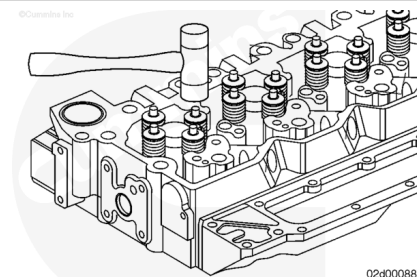
Remove the Cummins® valve spring compressor service tool, Part Number 3164329.



⚠ WARNING ⚠

To reduce the possibility of personal injury, wear eye protection. If the collets are not correctly installed, they can fly out when the stems are hit with a hammer.

After assembly, hit the valve stems with a plastic hammer to make sure the collets are seated.



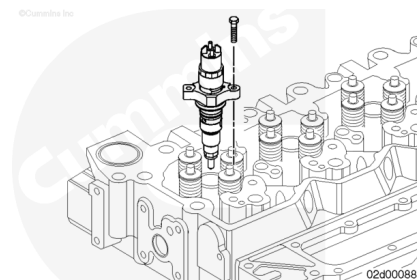
Measure

⚠ CAUTION ⚠

Improper injector protrusion can cause performance problems and high-pressure fuel leaks due to misalignment of the fuel connector.

Install the injectors with sealing washers into the cylinder head. Refer to Procedure 006-026 in Section 6.
 (/qs3/pubsys2/xml/en/procedures/216/216-006-026-tr.html)

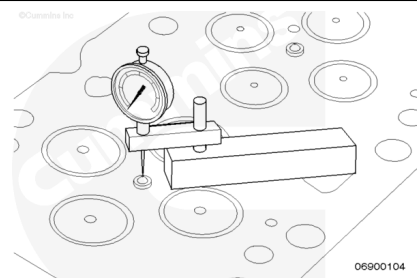
Measure the injector protrusion.



Install Cummins® depth gauge assembly, Part Number 3164438, on the cylinder head combustion deck and zero.

Rotate the depth gauge so it is measuring the injector protrusion at the highest point on the injector.

Measure the injector protrusion for each injector.

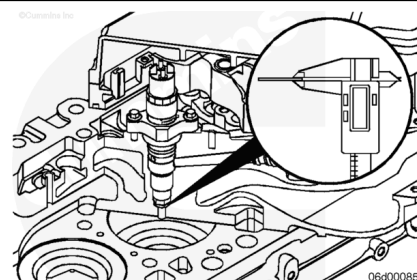


Injector Protrusion		
mm		in
2.45	MIN	0.096
3.15	MAX	0.124

If the injector protrusion is out of specification, check the thickness of the injector sealing washer. Refer to Procedure 006-026 in Section 6.

(/qs3/pubsys2/xml/en/procedures/216/216-006-026-tr.html)

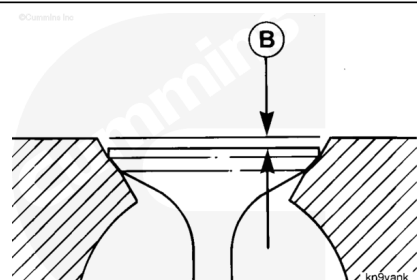
If the sealing washer is the correct thickness, check to make sure the injector bore is clean and free of debris. Also make sure that the sealing washers are **not** 'stacked' in the injector bore.



Install Cummins® depth gauge assembly, Part Number 3164438, on the cylinder head combustion deck and zero.

Rotate the depth gauge so it is measuring the valve recession into the cylinder head (B).

Measure the valve depth for each valve.



Intake Valve Depth (Installed)		
mm		in
0.584	MIN	0.023
1.092	MAX	0.043

Exhaust Valve Depth (Installed)

mm		in
0.965	MIN	0.038
1.473	MAX	0.058

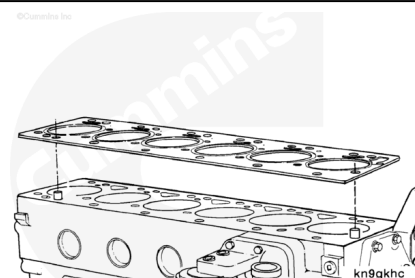
If the valve depth is **not** within specification, check if debris is preventing the valve from closing completely. If no debris is found, remove the valve and inspect the valve seat and valve face for damage.

Install

⚠ CAUTION ⚠

Make sure the gasket is correctly aligned with the holes in the cylinder block. Damage to the cylinder block can occur if the gasket is not aligned correctly.

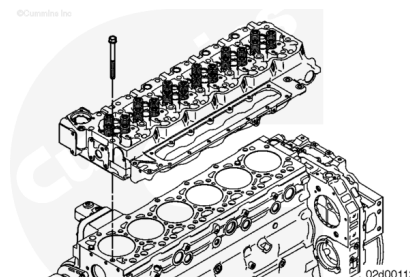
Install the cylinder head gasket. Refer to Procedure 002-021 in Section 2. (</qs3/pubsys2/xml/en/procedures/220/220-002-021-tr.html>)



⚠ WARNING ⚠

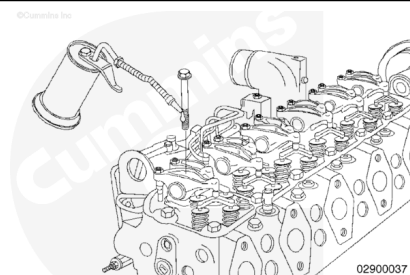
This component or assembly weighs greater than 23 kg [50 lb]. To reduce the possibility of serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.

Carefully put the cylinder head on the cylinder block, and seat it onto the dowels.



Lubricate the threads and under the heads on the cylinder head mounting capscrews with clean engine oil.

Install the capscrews and tighten hand-tight.

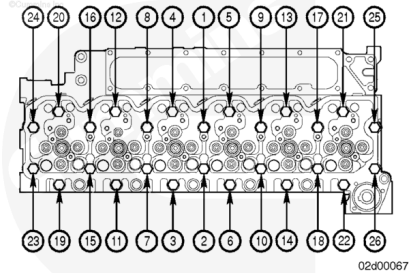


Use the illustrated sequence to tighten the cylinder head capscrews.

Tighten the cylinder head capscrews.

Torque Value:

1. 90 n•m [66 ft-lb]
2. Loosen and tighten again.
3. 90 n•m [66 ft-lb]
4. Rotate 90-degrees clockwise.



Finishing Steps

⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

⚠ CAUTION ⚠

If the cylinder head was installed with the injectors installed, it may be necessary to loosen the injector to correctly install the fuel connectors. Failure to install the fuel connectors will result in excessive injector fuel leakage and poor engine performance.



- Install the fuel connectors. Refer to Procedure 006-052 in Section 6. (</qs3/pubsys2/xml/en/procedures/216/216-006-052-tr.html>)
- Install the fuel rail to the cylinder head. Refer to Procedure 006-060 in Section 6. (</qs3/pubsys2/xml/en/procedures/216/216-006-060-tr.html>)
- Install the injector supply lines. Refer to Procedure 006-051 in Section 6. (</qs3/pubsys2/xml/en/procedures/216/216-006-051-tr.html>)
- Install the fuel drain lines. Refer to Procedure 006-013 in Section 6. (</qs3/pubsys2/xml/en/procedures/216/216-006-013-tr.html>)
- Install the push rods. Refer to Procedure 004-014 in Section 4. (</qs3/pubsys2/xml/en/procedures/220/220-004-014-tr.html>)
- Install the rocker levers. Refer to Procedure 003-008 in Section 3. (</qs3/pubsys2/xml/en/procedures/220/220-003-008-tr.html>)
- Install the crossheads. Refer to Procedure 002-001 in Section 2. (</qs3/pubsys2/xml/en/procedures/220/220-002-001-tr.html>)
- Adjust the overhead. Refer to Procedure 003-004 in Section 3. (</qs3/pubsys2/xml/en/procedures/216/216-003-004-tr.html>)
- If equipped, install the rocker lever housing. Refer to Procedure 003-013 in Section 3. (</qs3/pubsys2/xml/en/procedures/216/216-003-013-tr.html>)
- Install the rocker lever cover gasket and rocker lever cover. Refer to Procedure 003-011 in Section 3. (</qs3/pubsys2/xml/en/procedures/216/216-003-011-tr.html>)
- Install the exhaust manifold. Refer to Procedure 011-007 in Section 11. (</qs3/pubsys2/xml/en/procedures/216/216-011-007-tr.html>)
- Install the turbocharger. Refer to Procedure 010-033 in Section 10. (</qs3/pubsys2/xml/en/procedures/216/216-010-033-tr.html>)
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Note : Omit the following steps if the engine is not equipped with the component or if it was not necessary to remove the component to remove the cylinder head.

- Install the coolant temperature sensor. Refer to Procedure 019-019 in Section 19.
(/qs3/pubsys2/xml/en/procedures/216/216-019-019.html)
- Install the thermostat housing and thermostat from the engine. Refer to Procedure 008-013 in Section 8.
(/qs3/pubsys2/xml/en/procedures/216/216-008-013-tr.html)
- Install the alternator bracket to the thermostat housing. Refer to Procedure 013-003 in Section 13.
(/qs3/pubsys2/xml/en/procedures/216/216-013-003-tr.html)
- Pivot the alternator toward the engine. Tighten the alternator link, mounting bolt, and water inlet connection capscrews. Refer to Procedure 013-001 in Section 13.
(/qs3/pubsys2/xml/en/procedures/216/216-013-001-tr.html)
- Install the fan hub assembly. Refer to Procedure 008-036 in Section 8. (/qs3/pubsys2/xml/en/procedures/216/216-008-036-tr.html)
- Install the fan hub pulley. Refer to Procedure 008-039 in Section 8. (/qs3/pubsys2/xml/en/procedures/216/216-008-039-tr.html)
- Install the drive belt. Refer to Procedure 008-002 in Section 8. (/qs3/pubsys2/xml/en/procedures/220/220-008-002-tr.html)
- Install the air intake manifold. Refer to Procedure 010-023 in Section 10.
(/qs3/pubsys2/xml/en/procedures/216/216-010-023-tr.html)
- Install the fuel block connection. Refer to Procedure 006-021 in Section 6.
(/qs3/pubsys2/xml/en/procedures/216/216-006-021-tr.html)
- Install the fuel filter head bracket. Refer to Procedure 006-018 in Section 6.
(/qs3/pubsys2/xml/en/procedures/216/216-006-018-tr.html)
- Install the fuel filter and fuel filter head. Refer to Procedure 006-017 in Section 6.
(/qs3/pubsys2/xml/en/procedures/216/216-006-017-tr.html)

Note : The following components are required to be installed in order to complete the installation of the cylinder head.

- Connect the air crossover connection. Refer to Procedure 010-019 in Section 10. (</qs3/pubsys2/xml/en/procedures/220/220-010-019.html>)
- Connect any necessary engine wiring harness connections.
- Connect all water and heater hoses attached to the cylinder head. See the equipment manufacturer service information.
- Install any OEM accessories attached to the cylinder head. See the equipment manufacturer service information.
- Fill the engine with coolant. Refer to Procedure 008-018 in Section 8. (</qs3/pubsys2/xml/en/procedures/220/220-008-018-tr.html>)
- Connect the batteries. See the equipment manufacturer service information.
- Prime the fuel system. Refer to Procedure 006-015 in Section 6. (</qs3/pubsys2/xml/en/procedures/216/216-006-015-tr.html>)
- Operate the engine and check for leaks.